Applications of Distributed Learning to Large Language Models

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1 Motivations

The applications of distributed learning to large language models (LLMs) are mainly focused on the fine-tuning of LLMs. There are two main reasons for this situation: First, with the wide application of LLMs in daily life, the demand for domain-specific LLMs is growing quickly. Each general-purpose LLM, such as GPT-4, LLaMA, Qweh and Mistra, can be derived from various domain-specific LLMs. Second, training a general-purpose LLM requires a lot of computing power, which is currently available to only a few companies. At this stage, the main goal of these companies is to build efficient computing clusters rather than distributed training large models, as distributed learning is generally less efficient than centralized learning.

1.1 Split Learning in LLMs

Many institutions and organizations, including enterprises, academic institutions, and government agencies, run out of computing power when fine-tuning LLMs for their industry. They often need to rely on the computing power provided by public servers to address this issue. However, for privacy reasons, these entities cannot upload raw data directly to public servers. Split learning (SL) can mitigate privacy risks by preprocessing and desensitizing private data in the client model before it is sent to a public server.

1.2 Federated Learning in LLMs

Building domain-specific LLMs often requires fine-tuning of sensitive data from different institutions and organizations, but due to privacy concerns, they cannot directly share sensitive data to build fine-tuning datasets. Federated learning (FL) is an effective solution in this scenario, keeping sensitive data secure as they fine-tune LLMs.

In the scenario of fine-tuning LLMs, the clients (edge nodes) in FL are no longer small mobile devices but rather computing clusters owned by different entities. Some studies have explored the possibility of training LLMs on small mobile devices, arguing that modern smartphones are starting to include built-in GPU and NPU components. However, this remains purely theoretical, with no experimental validation.

2 Related Work

In studies combining distributed learning and LLMs, theoretical analysis often differs from experimental settings. Specifically, while the theory examines the impact of distributed learning on LLMs, experiments are frequently conducted using the ViT model. This section focuses on ensuring the use of LLMs in both theoretical analysis and experiments.

The application of distributed learning to LLMs primarily addresses two key aspects: privacy and efficiency. Table 1 presents key information from several papers on the combination of FL and SL with LLMs. Additional details from these articles are discussed in Sections 1, 3, and 4.

Paper	FL	SL	Research Direction	Training Mode	Full parameter or LoRA
[1]		✓	Privacy	GPT-2, OPT, Llama, and Qwen	Both
[2]	✓		Privacy and efficiency	/	Both
[3]		✓	Efficiency	LLaMA2-7B	Full parameter
[4]		✓	Efficiency	GPT-2	LoRA
[5]		✓	Efficiency	BERT-Base	LoRA
[6]	✓		Privacy	LlaMa2-7B	LoRA
[7]	✓		Efficiency	LLaMA-3	LoRA
[8]	✓		Efficiency	/	/

Table 1: Related work of applications of distributed learning to LLMs

Moldel	Mode	GPU
GPT-neo-2.7 B	LoRA	1 L40
GPT-neo-2.7 B	Full Parameters	/
Llama-7B	LoRA	8 A100
GPT2-M	LoRA	1 3090
GPT2-S	LoRA	1 3090
LLaMA-3-8B	LoRA	16 A100
qwen2.5-0.5B	LoRA	4 4090
qwen2.5-0.5B	Full Parameters	1 4090

Table 2: LLMs and GPUs required for training

3 Potential Directions

After investigation, I believe that applying split federatred learning (SFL) to LLMs is a highly promising direction. The reasons are as follows. First, both FL and SL have inherent limitations when applied to LLMs: FL demands significant computational resources from each client, while SL cannot use the data of multiple clients. Second, this area remains unexplored, and developing a comprehensive framework that integrates split models, training strategies, and deployment mechanisms would represent a significant contribution. Third, fine-tuning with SFL outperforms FL because SFL can leverage large-scale computing clusters without reducing the number of trainable parameters, whereas existing FL frameworks minimize trainable parameters as much as possible. Fourth, training LLMs is highly time-consuming, usually requiring at least a few hours per epoch. In distributed learning, the overall bottleneck was previously a combination of training time and communication time. However, when training LLMs, training time becomes the dominant limiting factor, and communication time is no longer the bottleneck. This gives SFL an advantage in practical deployments, as it involves more communication than FL. In contrast, for image classification tasks, the server's computing advantage over devices like the Raspberry Pi is less significant, and SFL generates much more data traffic than FL, resulting in a slower training speed in real-world deployment.

4 Experiment

I ran some experiments this week, the main purpose of which was to learn to train LLMs and screen the LLMs that the current lab's servers can run. I successfully fine-tuned GPT-neo-2.7 B using LoRA, and this model is of moderate size, making it suitable as a baseline for subsequent work. Meanwhile, Table 2 shows the GPUs required by different LLMs.

References

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